The worker is the constant-branch slice of the weld

A bicomodule comparison of state-entanglement and data-aggregation

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Abstract

In the polynomial-functor semantics of stateful agents, two constructions package the state of a worker: the *weld* $B = \mathsf{St}S \triangleleft \mathsf{St}T$, which composes two costate comonads, and the *worker grading* $A = \mathsf{St}(S \times T)$, which tensors them. It was recently established that B is a genuine $(\mathsf{St}T, \mathsf{St}S)$-bicomodule whose represented parametric-right-adjoint functor is $\bar{F}_B(V) = (T \times V)^S$. The status of the worker A inside this bicomodule was left open, and a natural conjecture — that A is the right-adjoint "content" factor of B — had resisted a clean verdict, because A and B live at different levels of the framed bicategory $\mathsf{Cat}^\sharp$ and so cannot be compared as written. We resolve this by reading A in the same variance as B: the worker is canonically the *constant-branch sub-bicomodule* $B' \subseteq B$, with carrier polynomial exactly A. Worker and weld then *share* the content power $(-)^S$ and differ *only* in their coproduct-aggregation index — the constant branch maps T for the worker against all branch maps T^S for the weld. The separation is proper precisely on $|S|, |T| \geq 2$, and degenerates (worker = weld on the nose) when $|S| = 1$ or $|T| = 1$. State-entanglement and data-aggregation are thus genuinely different collapses of one bicomodule.

1 Introduction

The problem. A stateful agent — a *worker* — is modelled in the container/polynomial program as an interface together with a store of internal state. When a worker with input set S and output set T carries its state, two polynomial constructions compete to express "state over the interface". The first is the *weld*

$$B \;=\; \mathsf{St}S \triangleleft \mathsf{St}T,$$

the composition-product ($\triangleleft$) of the costate comonads of S and T; this is the Zappa–Szép style orchestration weld, in which S runs and then T runs. The second is the *worker grading*

$$A \;=\; \mathsf{St}S \otimes \mathsf{St}T \;=\; \mathsf{St}(S \times T),$$

the Dirichlet tensor of the two comonads, which fuses the two states into a single store on $S \times T$. Both are comonoids — small categories — in $(\mathsf{Poly}, \triangleleft, \mathsf{y})$, and both claim to be "the state of the composite worker". Which is right, and how are they related?

The gap. The question acquired teeth once the weld was placed inside a framed bicategory. By the Spivak–Garner–Fairbanks correspondence [1], the framed bicategory $\mathsf{Cat}^\sharp$ of categories and parametric right adjoints is equivalent to $\mathsf{Comod}(\mathsf{Poly})$, so a horizontal 1-cell $c \to d$ is a *bicomodule*, equivalently a parametric right adjoint (PRA) functor $c\text{-}\mathsf{Set} \to d\text{-}\mathsf{Set}$. A companion note established three facts about the weld, of which the first was independently refereed and confirmed:

(1) *(existence, peer-reviewed)* $B = \mathsf{St}S \triangleleft \mathsf{St}T$ carries a genuine $(\mathsf{St}T, \mathsf{St}S)$-bicomodule structure, hence is a PRA functor $\mathsf{St}T\text{-}\mathsf{Set} \to \mathsf{St}S\text{-}\mathsf{Set}$, with

$$\bar{F}_B(V) \;\cong\; (T \times V)^S \;=\; T^S \times V^S;$$

(2) *(identification)* the worker A is the right-adjoint "content" factor of B under the SGF slice factorization;

(3) *(tail)* the data discarded by the worker's retract is the coproduct aggregation tail of B.

Clauses (2) and (3) were the live question, and the obstacle was a *type error*: the worker A is a priori a 0-cell of $\mathsf{Cat}^\sharp$ (a comonoid, a category), while the weld content is a 1-cell (a bicomodule $\mathsf{St}T \to \mathsf{St}S$). One cannot ask whether a category "equals" the content of a bicomodule until one has read the category as a bicomodule of the same variance. Until that reading is fixed, any comparison of exponents is comparing objects at different levels. A second, subtler error compounds this one: inferring from the carrier coincidence $B = A$ at $|S| = 1$ that the worker must there be the content factor. Equal carriers do not entail equal content factors — at $(|S|, |T|) = (1, m)$ with $m \geq 2$ the carriers coincide, yet the worker is *not* the content factor — so the coincidence $B = A$ settles nothing about content.

The result. We supply the missing reading and settle (2) and (3) together. The worker has a canonical $(\mathsf{St}T, \mathsf{St}S)$-bicomodule structure *inside* the weld: it is the sub-bicomodule carried by the *constant* branch maps.

Theorem 1 (Worker = constant-branch slice; see Proposition 1 and Theorems 5, 6)**.** *Let S, T be finite nonempty sets. Read the worker $A = \mathsf{St}(S \times T)$ as a $(\mathsf{St}T, \mathsf{St}S)$-bicomodule via the inclusion $\sigma\colon A \xrightarrow{\sim} B'$ onto the constant-branch sub-bicomodule $B' \subseteq B$ ($\sigma_1(s,t) = (s, \mathrm{const}_t)$, $\sigma^\sharp = \mathrm{id}_{S\times T}$). Then*

$$\bar{F}_A(V) \;=\; \textstyle\sum_{g \in \mathrm{const}(T)} V^S \;=\; T \cdot V^S \qquad \subseteq \qquad \bar{F}_B(V) \;=\; \textstyle\sum_{g \in T^S} V^S \;=\; T^S \cdot V^S.$$

Worker and weld share *the right-adjoint content power $(-)^S$; they differ* only *in the coproduct-aggregation index, $T \cong \mathrm{const}(T)$ for the worker against all branch maps $T^S = \mathrm{Hom}(S, T)$ for the weld. The inclusion $\bar{F}_A \hookrightarrow \bar{F}_B$ is an equality iff $|S| = 1$ or $|T| = 1$, and is proper precisely on $|S| \geq 2 \wedge |T| \geq 2$.*

Thus clause (2) is **false** where it is interesting ($|S|, |T| \geq 2$) — but *not* for the reason first proposed. The content *power* $(-)^S$ is in fact *shared*; the worker is a *sub-coproduct* of the weld's content, T copies against T^S. We are careful to separate two readings that the companion note conflated: the worker is the bare *content factor* $(-)^S$ only when $|T| = 1$, whereas it equals the *whole* weld (carriers $A = B$) when $|S| = 1$ or $|T| = 1$ — so at $(1, 2)$ the worker is the whole weld yet not the content factor. And clause (3), once stated honestly, splits into a true half (the aggregation tail *is* the full branch-map set T^S) and a false half (the retract does *not* discard "the non-constant maps"); moreover the set-level collapse e realising it is not itself a bicomodule map — the genuine bicomodule retraction is a base-pointed variant r_{s_0}.

The method. The one idea is Definition 3: inside the weld's position set $S \times T^S$, the positions with *constant* branch map form a sub-polynomial whose carrier is exactly A, and this slice is coaction-closed because it is S-*saturated* (Lemma 3(b)): λ moves the input over all of S while leaving the branch map untouched, so a set closed under that motion and built from constant branch maps stays inside the constant slice. Everything else is bookkeeping on the peer-reviewed formula $\bar{F}_B(V) = (T \times V)^S$: restrict its coproduct index from T^S to $\mathrm{const}(T)$ (Section 5), and read off what the worker's retract keeps and discards (Section 6).

Consequences. The two constructions carry a genuine semantic distinction that the orchestration calculus must respect. State-*entanglement* (the worker A) records one branch choice per output token $t \in T$; data-*aggregation* (the weld B) records one branch choice per response policy $g \in T^S$. Both read the whole input state — the shared content $(-)^S$ — but the weld aggregates over all policies and the worker only over the constant ones. The exact locus where the distinction bites is $|S|, |T| \geq 2$: with a single input or a single output there is nothing to entangle, and $A = B$.

Status and outline. This is a referee-response revision: clause (1) and the formula $\bar{F}_B(V) = (T \times V)^S$ are peer-reviewed input, used but not re-litigated. Section 2 fixes conventions (bicomodules, their handedness, the PRA correspondence, the costate comonad). Section 3 recalls the clause-(1) core. Section 4 proves the new content: the worker is the constant-branch subbicomodule. Section 5 carries out the corrected comparison, Section 6 the corrected retract analysis. Section 7 states the unified result and its orchestration reading. Section 8 is a point-by-point response to the referee.

2 Preliminaries

We work in the category Poly of polynomial functors $\mathsf{Set} \to \mathsf{Set}$ and natural transformations, embedded fully faithfully in $[\mathsf{Set}, \mathsf{Set}]$ [1, 2]. A polynomial $p = \sum_{i \in p(1)} \mathsf{y}^{p[i]}$ has a set of *positions* $p(1)$ and, at each position i, a *direction set* $p[i]$; it acts by $p(X) = \sum_i X^{p[i]}$. The *composition product* is functor composition $(p \triangleleft q)(X) = p(q(X))$ with unit y; the *Dirichlet product* $\otimes$ has positions $p(1) \times q(1)$, directions $p[i] \times q[j]$, and unit y. A morphism $\phi \colon p \to q$ is a function $\phi_1 \colon p(1) \to q(1)$ on positions together with a backward function $\phi^\sharp \colon q[\phi_1 i] \to p[i]$ on directions, for each i.

Comonoids are categories. A comonoid in $(\mathsf{Poly}, \triangleleft, \mathsf{y})$ is exactly a small category [1, Prop. 5.2]: the positions are objects, the directions at an object are the morphisms out of it, the counit selects identities and the comultiplication is composition. We write $\mathsf{Cat}^\sharp$ for the framed bicategory whose objects are these polynomial comonoids.

Definition 1 (Bicomodule; [1, Def. 4.2, Rmk 4.3])**.** Let $(c, \varepsilon_c, \delta_c)$ and $(d, \varepsilon_d, \delta_d)$ be comonoids in $(\mathsf{Poly}, \triangleleft, \mathsf{y})$. A (c, d)-*bicomodule* $m \colon c \to d$ is a carrier $m \in \mathsf{Poly}$ with a left d-coaction and a right c-coaction

$$d \triangleleft m \xleftarrow{\lambda} m \xrightarrow{\rho} m \triangleleft c$$

satisfying counit, coassociativity, and compatibility laws. (The variance follows the Leibniz reading of [1, Rmk 4.3]: λ sends m to "d after m" and ρ to "m after c", so m goes from c to d.)

Theorem 2 (SGF correspondence; [1, Prop. 3.9, Thm. 4.28])**.** *For comonoids c, d, the framed bicategories $\mathsf{Cat}^\sharp$ and $\mathsf{Comod}(\mathsf{Poly})$ are equivalent; in particular*

$$\mathsf{Comod}(\mathsf{Poly})(c, d) \;\simeq\; \mathsf{PRA}\big(c\text{-}\mathsf{Set},\ d\text{-}\mathsf{Set}\big),$$

so a (c,d)-bicomodule is *a parametric right adjoint functor c-$\mathsf{Set} \to d$-Set. For a c-variable polynomial F, the SGF slice factorization writes the PRA as a coproduct of representables indexed by $F(1)$, with right-adjoint content given by the connected-limit part [1, Prop. 3.6].*

Definition 2 (Costate comonad; [1, Ex. 5.6])**.** For a nonempty set S write

$$\mathsf{St}S \;=\; S \cdot \mathsf{y}^S \;=\; \left[\begin{smallmatrix} S \\ S \end{smallmatrix}\right],$$

the *costate* (or store) comonad, i.e. the comonad induced by the adjunction $S\times(-) \dashv (-)^S \colon \mathsf{Set} \to \mathsf{Set}$. It has S positions, each with direction set S. As a category it is the *indiscrete category* $\mathbb{I}(S)$ on object set S: a unique morphism between any two objects.

Throughout, S, T are finite and nonempty. We reserve the symbol Δ for the diagonal functor and never for a comonoid; the two costate comonoids are $\mathsf{St}S$ and $\mathsf{St}T$, and the worker carrier is $\mathsf{St}(S \times T)$.

Lemma 1 (The indiscrete collapse)**.** *For $S \neq \emptyset$ the indiscrete category $\mathbb{I}(S)$ is equivalent to the terminal category $\mathbf{1}$; hence $\mathsf{St}S$-$\mathsf{Set} \simeq \mathsf{Set}$, and under this equivalence every copresheaf X has $\lim X \cong X(s_0) \cong \operatorname{colim} X$ for each object s_0, and the representable at any object is the terminal copresheaf.*

Proof. In $\mathbb{I}(S)$ there is a unique isomorphism between any two objects, so the inclusion $\mathbf{1} \to \mathbb{I}(S)$ of any object is an equivalence. Equivalent categories have equivalent copresheaf categories, and limits/colimits transport accordingly. □

We use $\bar{F}$ for the endofunctor of Set obtained by transporting a store-bicomodule's PRA functor across the collapses $\mathsf{St}S$-$\mathsf{Set} \simeq \mathsf{Set}$, $\mathsf{St}T$-$\mathsf{Set} \simeq \mathsf{Set}$; right-adjointness and the SGF factorization are equivalence-invariant.

3 The weld bicomodule (peer-reviewed core)

We recall the clause-(1) results, which are input to what follows. The naming departs from the companion note in one respect: $B = \mathsf{St}S \triangleleft \mathsf{St}T$ is *not* a regular bicomodule (the regular bicomodule of a comonoid c is c as a (c,c)-bicomodule). It is the *unit-composite* bicomodule — the composite over the unit y of the two one-sided regular comodules — as Theorem 4 makes explicit.

Lemma 2 (Unit-composite bicomodule)**.** *For any comonoids c, d in a monoidal category $(\mathcal{V}, \triangleleft, \mathsf{y})$, the object $d \triangleleft c$ is a (c,d)-bicomodule with*

$$\lambda \;=\; \delta_d \triangleleft c \colon d \triangleleft c \to d \triangleleft d \triangleleft c, \qquad \rho \;=\; d \triangleleft \delta_c \colon d \triangleleft c \to d \triangleleft c \triangleleft c.$$

Proof. The left laws (counit, coassociativity) are the comonoid laws of d whiskered on the right by $\triangleleft c$; the right laws are the comonoid laws of c whiskered on the left by $d\triangleleft$. For compatibility, λ acts on the first (d) factor and ρ on the last (c) factor; since $\triangleleft$ is a bifunctor, the two whiskerings commute and both composites equal $\delta_d \triangleleft \delta_c$. □

Theorem 3 (Clause (1), peer-reviewed)**.** *The weld $B = \mathsf{St}S \triangleleft \mathsf{St}T$ is the unit-composite $(\mathsf{St}T, \mathsf{St}S)$-bicomodule (take $c = \mathsf{St}T$, $d = \mathsf{St}S$, carrier $d\triangleleft c = \mathsf{St}S\triangleleft\mathsf{St}T$), with $\lambda = \delta_{\mathsf{St}S}\triangleleft\mathsf{St}T$ and $\rho = \mathsf{St}S \triangleleft \delta_{\mathsf{St}T}$. Hence B is a horizontal 1-cell of $\mathsf{Cat}^\sharp$, i.e. a PRA functor $\mathsf{St}T$-$\mathsf{Set} \to \mathsf{St}S$-$\mathsf{Set}$.*

Proof. Lemma 2 with $c = \mathsf{St}T$, $d = \mathsf{St}S$; both are comonoids, and Theorem 2 supplies the PRA reading. □

Theorem 4 (The represented prafunctor; peer-reviewed)**.** *After the collapse of Lemma 1,*

$$\bar{F}_B(V) \;\cong\; (T \times V)^S \;=\; T^S \times V^S.$$

As a polynomial endofunctor of Set*,* $\bar{F}_B = T^S \cdot \mathsf{y}^S$*: its position set (the coproduct index* $\bar{F}_B(1)$*) is the branch-map set* $T^S = \mathrm{Hom}(S, T)$*, and its direction set is* S*.*

Proof. The unit-composite factors over y as

$$\mathsf{St}T \xrightarrow{G} \mathsf{y} \xrightarrow{H} \mathsf{St}S, \qquad H \triangleleft G \;=\; \mathsf{St}S \triangleleft \mathsf{St}T \;=\; B,$$

where G is the right $\mathsf{St}T$-comodule with carrier $\mathsf{St}T$ and $\rho = \delta_{\mathsf{St}T}$, and H is the left $\mathsf{St}S$-comodule with carrier $\mathsf{St}S$ and $\lambda = \delta_{\mathsf{St}S}$. Bicomodule composition over y is just $\triangleleft$, so $F_B = F_H \circ F_G$.

For $F_G \colon \mathsf{St}T\text{-}\mathsf{Set} \to \mathsf{Set}$, the right $\mathsf{St}T$-comodule with carrier $\mathsf{St}T$ is

$$F_G(X) \;=\; \sum_{t \in T} \mathsf{St}T\text{-}\mathsf{Set}(\mathsf{y}_t, X),$$

where y_t denotes the representable at the object t. By Lemma 1 each y_t is terminal, so the hom-set is $\lim X$; hence $F_G(X) = \sum_{t \in T} \lim X$ and, after transport, $\bar{G}(V) = T \times V$.

For $F_H \colon \mathsf{Set} \to \mathsf{St}S\text{-}\mathsf{Set}$, a left $\mathsf{St}S$-comodule is a diagram $\mathsf{St}S \to \mathsf{Poly}$ [1, Prop. 5.7]; the regular one assigns to each object $a \in S$ the component polynomial y^S (one position, S directions, read off $\delta_{\mathsf{St}S}$). So $F_H(W)(a) = (\mathsf{y}^S)(W) = W^S$, and after transport $\bar{H}(W) = W^S$.

Therefore $\bar{F}_B(V) = \bar{H}(\bar{G}(V)) = (T \times V)^S$, and $(T \times V)^S = \{S \to T \times V\} = \{(g, h) : g \colon S \to T,\ h \colon S \to V\} = T^S \times V^S$. □

By Theorem 2 the SGF slice factorization of $\bar{F}_B(V) = \sum_{g \in T^S} V^S$ has coproduct ("Σ") index $\bar{F}_B(1) = T^S$ and right-adjoint *content* the power $(-)^S$ (represented by the direction set S). This content split is the backbone of the comparison below.

Remark 1. The represented prafunctor $\bar{F}_B = T^S \mathsf{y}^S$ is neither the underlying polynomial of $A = (S \times T)\mathsf{y}^{S \times T}$ nor that of the weld carrier $B = (S \cdot T^S)\mathsf{y}^{S \times T}$. The PRA functor of a bicomodule is not the underlying functor of its carrier — the trap at the heart of the problem. (At $|S| = |T| = 2$, $V = 2$: $\bar{F}_B = 16$, $A_{\mathrm{und}} = 64$, $B_{\mathrm{und}} = 128$.)

4 The worker as a bicomodule: the constant-branch slice

Here is the new content. To compare the worker A with the weld content we must read A as a $(\mathsf{St}T, \mathsf{St}S)$-bicomodule of the *same* variance as B. There is a canonical such reading, and it sits visibly inside B.

Recall the weld carrier as a polynomial: $(\mathsf{St}S \triangleleft \mathsf{St}T)(X) = S \times (T \times X^T)^S = S \times T^S \times X^{S \times T}$, so B has position set $S \times T^S$ — a current input $s \in S$ and a branch map $g \colon S \to T$ — and direction set $S \times T$ at each position.

Definition 3 (Constant-branch slice)**.** Let $\mathrm{const} \colon T \hookrightarrow T^S$, $t \mapsto \mathrm{const}_t$, be the inclusion of the constant branch maps. Let $B' \subseteq B$ be the sub-polynomial on the positions with constant branch map,

$$B' \;=\; \big\{(s, g) \;:\; g \in \mathrm{const}(T)\big\},$$

with inherited direction set $S \times T$. Its position set is $S \times \mathrm{const}(T) \cong S \times T$, so as a polynomial

$$B' \;=\; (S \times T) \cdot \mathsf{y}^{S \times T} \;=\; \mathsf{St}(S \times T) \;=\; A$$

on the nose.

Lemma 3 (Branch decomposition and the closed subobjects)**.** *For each branch map $g \in T^S$ let $B_g \subseteq B$ be the positions with that branch map, $B_g = \{(s, g) : s \in S\}$, a sub-polynomial with $|S|$ positions and directions $S \times T$. Then:*

(a) *each B_g is a $(\mathsf{St}T, \mathsf{St}S)$-sub-bicomodule of B, and $B = \coprod_{g \in T^S} B_g$ as bicomodules; all the B_g are mutually isomorphic;*

(b) *a sub-polynomial of B is coaction-closed iff it is S-*saturated*, i.e. a union $\coprod_{g \in G} B_g$ for some $G \subseteq T^S$;*

(c) *$B' = \coprod_{g \in \mathrm{const}(T)} B_g$ is coaction-closed; its carrier is A.*

Proof. The two coactions act on the position (s, g) as follows. The left coaction $\lambda = \delta_{\mathsf{St}S} \triangleleft \mathsf{St}T$ *moves the input* s over all of S while leaving the branch map g untouched: $\delta_{\mathsf{St}S}$ replicates the position across the input set, $\lambda(s, g) = \big(s,\ a \mapsto (a, g)\big)$, the inner copies ranging over $a \in S$ with the *same* g. The right coaction $\rho = \mathsf{St}S \triangleleft \delta_{\mathsf{St}T}$ acts on the outer $\mathsf{St}S$-copy and reorganizes the inner $\mathsf{St}T$-layer by $\delta_{\mathsf{St}T}$; it fixes the whole position (s, g), in particular g. (This is the corrected reading: g never appears as an argument of $\delta_{\mathsf{St}T}$, and it is λ, not ρ, that moves s.)

Hence a position set is closed under both coactions iff, whenever it contains (s, g), it contains (a, g) for every $a \in S$ — S-saturation — which is exactly (b); and each B_g (a single g, all inputs) is a minimal closed piece, giving the decomposition (a). The B_g are mutually isomorphic by relabelling g. For (c), $\mathrm{const}(T) \subseteq T^S$ yields the S-saturated union $B' = \coprod_{t \in T} B_{\mathrm{const}_t}$; it has $|T| \cdot |S|$ positions with directions $S \times T$, i.e. carrier $(S \times T) \cdot \mathsf{y}^{S \times T} = A$. The counit and (co)associativity laws restrict because they are the comonoid laws of $\mathsf{St}S, \mathsf{St}T$ whiskered, and B' is a full sub-on-positions with inherited directions. □

Remark 2 (what singles out B')*.* Closure does *not* single out B': by Lemma 3(b) every subset $G \subseteq T^S$ gives a closed sub-bicomodule, and there are $2^{|T^S|}$ of them. What makes $\mathrm{const}(T)$ canonical is *naturality*. The constant assignment is the diagonal $\Delta \colon T \to T^S = \mathrm{Hom}(S, T)$, $t \mapsto \mathrm{const}_t$, the unique map with $\mathrm{ev}_s \circ \Delta = \mathrm{id}_T$ for every input $s \in S$ — the unique natural (Yoneda) section of every evaluation. So among all S-saturated slices, B' is the one indexed by the canonical diagonal $T \hookrightarrow T^S$, and it is this that earns A its reading, not the fact (shared by many slices) of being closed.

Proposition 1 (The worker's canonical reading, via cobase change)**.** *Equip the worker carrier $A = \mathsf{St}(S \times T)$ with the $(\mathsf{St}T, \mathsf{St}S)$-bicomodule structure $\pi_! A$ obtained by* cobase change *of the regular (A, A)-bicomodule along the two comonoid projections $\pi_S \colon A \to \mathsf{St}S$ and $\pi_T \colon A \to \mathsf{St}T$ induced by the set projections of $S \times T$ (corestriction of the left coaction along π_S, of the right coaction along π_T). Then the carrier identity $\sigma \colon A \to B$, $\sigma_1(s, t) = (s, \mathrm{const}_t)$, $\sigma^\sharp = \mathrm{id}_{S \times T}$, is an isomorphism of $(\mathsf{St}T, \mathsf{St}S)$-bicomodules $\pi_! A \xrightarrow{\sim} B'$.*

Proof. Cobase change produces a genuine $(\mathsf{St}T, \mathsf{St}S)$-bicomodule on the carrier A with no reference to B, so the comparison is not circular: the left coaction is $A \xrightarrow{\delta_A} A \triangleleft A \xrightarrow{\pi_S \triangleleft A} \mathsf{St}S \triangleleft A$ and the right is $A \xrightarrow{\delta_A} A \triangleleft A \xrightarrow{A \triangleleft \pi_T} A \triangleleft \mathsf{St}T$. The two comonoid projections, as cofunctors, have direction parts $\pi_S{}^\sharp(a) = (a, t)$ and $\pi_T{}^\sharp(t') = (s, t')$ (a direction of the codomain is carried back to one of the domain, keeping the untouched coordinate fixed). Computing $\lambda_A = (\pi_S \triangleleft A)\, \delta_A$ and $\rho_A = (A \triangleleft \pi_T)\, \delta_A$ on positions and directions gives, in closed form for *all* S, T,

$$\begin{aligned} \lambda_A(s, t) &= \big(s,\ a \mapsto (a, t)\big), & \lambda_A{}^\sharp\big(a, (a', t')\big) &= (a', t'), \\ \rho_A(s, t) &= \big((s, t),\ (a, t') \mapsto t'\big), & \rho_A{}^\sharp\big((a, t'), t''\big) &= (a, t''). \end{aligned}$$

Now apply σ. On positions σ_1 is the bijection $S \times T \cong S \times \mathrm{const}(T)$ onto the B'-positions and $\sigma^\sharp = \mathrm{id}_{S \times T}$ matches directions, so pushing λ_A, ρ_A across σ reproduces exactly the restricted

weld coactions of Lemma 3 at $g = \mathrm{const}_t$: indeed $\lambda_B(s, \mathrm{const}_t) = \big(s,\ a \mapsto (a, \mathrm{const}_t)\big) = \big(s,\ a \mapsto \sigma_1(a,t)\big)$, and ρ_B fixes the whole position just as ρ_A does. Hence σ intertwines both coactions for all S, T, and a carrier bijection with $\sigma^\sharp = \mathrm{id}$ intertwining λ, ρ is an isomorphism of bicomodules onto its image B'. (The five bicomodule laws for $\pi_! A$ and both equivariance squares of σ were additionally machine-checked at all nine shapes $n, m \in \{1,2,3\}$, and $\bar{F}_A(V) = T \cdot V^S$ recomputed from A's own cobase-change structure, independently of σ.) □

5 The comparison: a coproduct-index separation

Theorem 5 (Corrected comparison)**.** *Read A and B as $(\mathsf{St}T, \mathsf{St}S)$-bicomodules via Section 4 ($A \cong B' \subseteq B$). Then*

$$\bar{F}_B(V) \;=\; \sum_{g \in T^S} V^S \;=\; T^S \cdot V^S, \qquad \bar{F}_A(V) \;=\; \sum_{g \in \mathrm{const}(T)} V^S \;=\; T \cdot V^S.$$

Both have the same *content power $(-)^S$ (direction set S); they differ* only *in the coproduct index, T^S for the weld against $T \cong \mathrm{const}(T)$ for the worker. Consequently $\bar{F}_A$ is a sub-prafunctor of $\bar{F}_B$, with $\bar{F}_A \cong \bar{F}_B$ iff $|S| = 1$ or $|T| = 1$, and a* proper *inclusion exactly on $|S| \geq 2 \wedge |T| \geq 2$.*

Proof. $\bar{F}_B = \sum_{g \in T^S} V^S$ is Theorem 4: the coproduct index is the weld positions modulo the left $\mathsf{St}S$-collapse (the S-head is killed by indiscreteness), namely the branch maps T^S, and the surviving direction is S (the T-direction is killed by the right $\mathsf{St}T$-collapse). By Proposition 1, $A \cong B'$ is B restricted on positions to $\mathrm{const}(T) \subseteq T^S$ with directions unchanged; the same recipe restricts the coproduct index to $\mathrm{const}(T)$ and leaves the content power $(-)^S$ untouched, giving $\bar{F}_A = \sum_{t \in T} V^S = T \cdot V^S$.

Equality of the two functors for all V is equality of polynomials $T^S \mathsf{y}^S$ and $T\mathsf{y}^S$, i.e. $|T^S| = |T|$, i.e. $m^n = m$ where $n = |S|, m = |T|$. For $n, m \geq 1$ this holds iff $n = 1$ or $m = 1$. Otherwise $T \subsetneq T^S$ and the inclusion $\bar{F}_A \hookrightarrow \bar{F}_B$ is proper. □

Remark 3 (what this corrects). The exponents *agree* — both content powers are $(-)^S$. An earlier argument found "S versus $S \times T$", but that compared B under its right $\mathsf{St}T$-coaction against A read with *no* right coaction (the y-reading). With the coaction present, the T-directions of A collapse into the exponent S, not $S \times T$. The consistent separation is the coproduct index T versus T^S, nonzero precisely where non-constant branch maps exist, i.e. on $|S| \geq 2 \wedge |T| \geq 2$. In particular at $|S| = 1$ one has $T^S = T$, so $\mathrm{const}(T) = T^S$ and $\bar{F}_A = \bar{F}_B = T \cdot V$; moreover the carriers coincide, $B = \mathsf{St}S \triangleleft \mathsf{St}T = \mathsf{y} \triangleleft \mathsf{St}T = \mathsf{St}T = \mathsf{St}(1 \times T) = A$. There the worker is the whole weld; it is the content factor only if also $|T| = 1$.

Verdict for clause (2). The original conjecture asks whether the worker is the right-adjoint *content factor* of the weld, i.e. whether $\bar{F}_A$ equals the content power $(-)^S$. We must not conflate this with the worker being the *whole* weld. Two distinct equalities are in play, and they have different loci:

- **worker = content factor:** $\bar{F}_A(V) = V^S$ (the bare content power), which holds *iff* $|T| = 1$ — a single output collapses the coproduct index T to a point;
- **worker = whole weld:** $\bar{F}_A = \bar{F}_B$ as prafunctors, equivalently the carriers coincide ($A = B$), which holds *iff* $|S| = 1$ *or* $|T| = 1$.

These are genuinely different: at $(|S|, |T|) = (1, 2)$ the carriers coincide, $A = B$ on the nose, yet $\bar{F}_A(V) = 2V \neq V =$ content, so the worker is the whole weld but *not* the content factor. Hence the conjecture "worker = content factor" is **true iff** $|T| = 1$ and **false whenever** $|T| \geq 2$; on the interesting locus $|S|, |T| \geq 2$ the worker is neither the content factor nor the whole weld. The content power $(-)^S$ is in all cases *shared*; what the worker is, is a sub-coproduct (T copies of the content against the weld's T^S).

Example 1. At $(|S|,|T|) = (2,2)$, $V = 2$: $\bar{F}_A = 2 \cdot 2^2 = 8$ and $\bar{F}_B = (2 \cdot 2)^2 = 16$ (proper). At $(1,2)$: $\bar{F}_A = \bar{F}_B = 4$; at $(1,3)$: $\bar{F}_A = \bar{F}_B = 6$ (equal). These are exactly the shapes where the criterion for worker $\neq$ content factor ("$|T| \geq 2$") and the criterion for worker $\neq$ weld ("$|S| \geq 2 \wedge |T| \geq 2$") disagree.

6 The retract: what the worker forgets

How does the weld project onto the worker slice B'? Here one must separate two levels, and the companion note conflated them. At the level of *position-sets* there is a clean idempotent collapse; at the level of *bicomodules* that collapse is not a morphism, and the genuine retraction is a different, base-pointed map. We record both honestly. Throughout, $n = |S|$, $m = |T|$, and the weld's coproduct index splits as

$$T^S \;=\; \mathrm{const}(T) \;\sqcup\; \mathrm{NC}, \qquad |\mathrm{const}(T)| = m, \quad |\mathrm{NC}| = m^n - m,$$

with NC the non-constant maps.

The set-level idempotent e. Let $e\colon S \times T^S \to S \times T^S$, $e(s,g) = (s, \mathrm{const}_{g(s)})$: collapse each branch map to the constant at its value on the *current* input s.

Theorem 6 (Keep/discard, as a map of position-sets)**.** *e is idempotent with image $S \times \mathrm{const}(T) \cong S \times T$, the worker positions $\sigma(A)$. For each current input s, e* keeps *the value $g(s) \in T$ (the diagonal T-coordinate) and* discards *the restriction $g|_{S\setminus\{s\}} \in T^{S\setminus\{s\}}$, a fibre of size m^{n-1}. Concretely the fibre of e over (s, const_t) is $\{g : g(s) = t\} \cong T^{S\setminus\{s\}}$.*

Proof. $e \circ e = e$ since $\mathrm{const}_{g(s)}$ evaluated at s is $g(s)$. The image is the constant slice $\{(s, \mathrm{const}_t)\}$. Two maps g, g' satisfy $e(s,g) = e(s,g')$ iff $g(s) = g'(s)$; fixing $g(s) = t$ leaves $S \setminus \{s\}$ free, so the fibre is $T^{S\setminus\{s\}}$, of size m^{n-1}. □

But e is not a bicomodule map. Theorem 6 is purely combinatorial; it does *not* make e a morphism of $(\mathsf{St}T, \mathsf{St}S)$-bicomodules, and in fact e is not one.

Proposition 2 (e fails λ-equivariance)**.** *With $e^\sharp = \mathrm{id}$ on directions, e satisfies the right (ρ-)equivariance square on all of B, but fails the left (λ-)equivariance square $\lambda \circ e = (\mathsf{St}S \triangleleft e) \circ \lambda$ at every position with a non-constant branch map; hence e is a bicomodule morphism iff every $g \in T^S$ is constant, i.e. iff $n = 1$ or $m = 1$. The first failure is at $(n,m) = (2,2)$, position $(s,g) = (0,(0,1))$.*

Reason. $\lambda = \delta_{\mathsf{St}S} \triangleleft \mathsf{St}T$ *moves the input* over all of S (Lemma 3). The map e reads g at the *current* input s; after λ has moved the input to some $a \neq s$, the composite $(\mathsf{St}S \triangleleft e) \circ \lambda$ broadcasts $g(a)$, whereas e applied first broadcasts $g(s)$ and λ merely copies it. On the witness $g = (0,1)$ the two give branch value $g(1) = 1$ against $g(0) = 0$ in the moved copy — unequal, so the square fails. (ρ fixes the whole position, so it never distinguishes the two orders.) A `decide`-checked verification over the whole carrier (ground set of size 2) confirms failure at $(2,2),(2,3),(3,2)$, and confirms the square holds on the boundary shapes $(1,2),(2,1)$ and on the slice B'. □

The genuine bicomodule retraction r_{s_0}. The fix is to read g at a *fixed* input, immune to λ's motion. Choose a base point $s_0 \in S$ and set $r_{s_0}(s,g) = (s, \mathrm{const}_{g(s_0)})$, again with $r^\sharp = \mathrm{id}$.

Theorem 7 (The retraction $B \twoheadrightarrow B'$)**.** *For any base point $s_0 \in S$, r_{s_0} is a morphism of $(\mathsf{St}T, \mathsf{St}S)$-bicomodules $B \to B'$ (it satisfies both the λ- and ρ-equivariance squares), is idempotent, has image the constant slice $B' = \sigma(A)$, and restricts to the identity on B' — a genuine bicomodule retraction. Its fibre over (s, const_t) is $\{g : g(s_0) = t\} \cong T^{S\setminus\{s_0\}}$, of size m^{n-1}: the*

same count *as e. Being a bicomodule map,* r_{s_0} *represents a genuine endomorphism of* $\bar{F}$*; the set-level e does not (Proposition 2), but e agrees with* r_{s_0} *on the fibre of* $\bar{F}$ *at the base object* s_0*, the point at which the collapse* $\mathsf{St}S\text{-}\mathsf{Set} \simeq \mathsf{Set}$ *is evaluated. The retraction* r_{s_0} *is honestly* non-canonical*: it depends on the choice of* s_0*.*

Proof. Because r_{s_0} reads g at the fixed s_0, both $r_{s_0} \circ \lambda$ and $\lambda \circ r_{s_0}$ broadcast $g(s_0)$, so the λ-square holds; ρ fixes the position and the ρ-square holds as for e. Idempotency, image, retraction, and the fibre count m^{n-1} are as in Theorem 6 with s replaced by s_0. All of λ- and ρ-equivariance, idempotency, image$= B'$, and $r_{s_0} \circ \iota_{B'} = \mathrm{id}$ were `decide`-checked at $(2,2),(2,3),(3,2),(1,2),(2,1)$ (and base-point independence at $s_0 \in \{0,1\}$ for $(2,2)$). □

So the bicomodule-level statement "the worker is the retract of the weld" is true with r_{s_0} in place of e; the set-level combinatorics (Theorem 6) are unaffected, and e agrees with r_{s_0} on the $\bar{F}$-fibre at s_0 (though only r_{s_0}, being a bicomodule map, represents an endomorphism of $\bar{F}$).

Re-decision of clause (3). As first stated — "the discarded data is precisely the non-constant branch maps, and that is the Σ-tail" — the claim is **false**, for two independent reasons.

(i) The Σ-tail index is *all* of $T^S = \mathrm{const}(T) \sqcup \mathrm{NC}$, not NC. "$T^S$ = the non-constant maps" is a cardinality contradiction ($m^n \neq m^n - m$).

(ii) What the retraction forgets per position is the fibre $T^{S\setminus\{s_0\}}$ of size m^{n-1} — neither NC (size $m^n - m$) nor T^S (size m^n). At $(2,3)$: 3 versus 6 versus 9, all distinct.

The **true residue** is this: the weld's Σ-tail *is* the full branch-map set $T^S = \mathrm{Hom}(S,T)$ — that half of clause (3) stands — and the retraction r_{s_0} realises the projection of this index onto the worker's constant slice $\mathrm{const}(T) \cong T$ by "read the branch at s_0, then make constant". The genuinely discarded structure is the off-diagonal dependence $g|_{S\setminus\{s_0\}}$. It is *non-trivial* — the retraction actually loses something, the fibre having size $m^{n-1} > 1$ — exactly on $|S| \geq 2 \wedge |T| \geq 2$, the same locus as the clause-(2) separation.

This also settles the old worry about a "reversal of adjointness" honestly: T^S is the index of the *left*-adjoint coproduct $\sum_{T^S} \dashv \Delta_{T^S}$, not right-adjoint content; the right-adjoint content $(-)^S$ is *kept* by both worker and weld; and the worker-versus-weld difference is purely how much of the coproduct index is retained (T versus T^S) — no adjoint-direction clash at all.

7 The unified statement

Theorem 8 (Worker = constant-branch slice of the weld)**.** *Let* S, T *be finite nonempty. With* A *read as the* $(\mathsf{St}T, \mathsf{St}S)$*-bicomodule* $\sigma(A) = B' \subseteq B$ *of constant branch maps* ($\sigma^\sharp = \mathrm{id}_{S\times T}$)*,*

$$\bar{F}_A(V) = T \cdot V^S \;\subseteq\; \bar{F}_B(V) = T^S \cdot V^S$$

is the sub-coproduct over $\mathrm{const}(T) \subseteq T^S$*. Worker and weld* share *the right-adjoint content power* $(-)^S$ *and differ* only *in the coproduct index (*T *versus* T^S*). For any base point* s_0*, the bicomodule retraction* $r_{s_0}(s,g) = (s, \mathrm{const}_{g(s_0)})$ *projects* T^S *onto* $\mathrm{const}(T)$*, forgetting the off-diagonal fibre* $T^{S\setminus\{s_0\}}$ *(size* m^{n-1}*); the set-level idempotent* $e(s,g) = (s, \mathrm{const}_{g(s)})$ *has the same image and fibre count but is not a bicomodule map. The degeneracies* $\bar{F}_A = \bar{F}_B$*, trivial discard, and* $A = B$ *coincide and hold* iff $|S| = 1$ or $|T| = 1$*; the proper separation is exactly* $|S| \geq 2 \wedge |T| \geq 2$*. (The worker is the bare content factor* $(-)^S$ *on the narrower locus* $|T| = 1$*.)*

Orchestration reading. State-*entanglement* (the worker A) keeps a branch choice per output token $t \in T$; data-*aggregation* (the weld B) keeps a branch choice per response policy $g \in T^S$. Both read the whole input state — the shared content $(-)^S$ — but the weld aggregates over all policies and the worker only over constant ones. Keeping the two distinct is a semantic requirement, not a defect, and the exact locus where it matters is $|S|, |T| \geq 2$: with a single input or output there is nothing to entangle.

Grant relevance. The weld now has a precise home in Poly: it is a data-migration prafunctor $\mathsf{St}T\text{-}\mathsf{Set} \to \mathsf{St}S\text{-}\mathsf{Set}$, $X \mapsto (T \times X)^S$, with the branch policy aggregated by $\sum_{T^S}$. This places the *state* axis of the compositional-correctness program inside the same framed bicategory $\mathsf{Cat}^\sharp$ that houses the directed (retrofunctor) axis. But the worker is not that prafunctor's content (for $|T| \geq 2$): entanglement and aggregation are different collapses of one bicomodule, and the orchestration semantics must keep them apart.

8 Response to the referee

This revision answers the referee's second- and third-round reports. The core finding — that B' is coaction-closed and is the worker's reading — was endorsed; the second round's two MUST items and three SHOULDs are addressed below, and the third round's residual items are folded into the same text. We record where each lands.

- **M1** (the retract e is not a bicomodule map): Section 6. Proposition 2 states that $e(s, g) = (s, \mathrm{const}_{g(s)})$ fails λ-equivariance at every non-constant g (first failure $(2, 2)$, position $(0, (0, 1))$), because e reads g at the *current* input while λ moves the input over S. Theorem 7 replaces it with the base-pointed $r_{s_0}(s, g) = (s, \mathrm{const}_{g(s_0)})$, a genuine bicomodule retraction $B \twoheadrightarrow B'$ (both squares), honestly non-canonical, keeping every count (m^{n-1} fibre) and the same $\bar{F}$. The set-level facts of Theorem 6 about e are unchanged; only the bicomodule-retraction reading switches $e \to r_{s_0}$. Both the failure of e and the retraction property of r_{s_0} were re-derived by the author (companion script; `decide`-checked in the Lean leaf).

- **M2** (clause-(2) verdict conflation): the verdict in Section 5 now separates the two readings and labels each. "Worker = content factor $(-)^S$" holds *iff* $|T| = 1$; "worker = whole weld" ($A = B$, equiv. $\bar{F}_A = \bar{F}_B$) holds *iff* $|S| = 1$ *or* $|T| = 1$; the counterexample $(1, 2)$ ($A = B$ yet $\bar{F}_A = 2V \neq V$) is spelled out. The phrase "content (indeed $A = B$)" is gone.

- **S1** (general sub-bicomodule fact; canonicity): Lemma 3 now gives $B = \coprod_{g \in T^S} B_g$ with all B_g mutually isomorphic, identifies the coaction-closed subobjects as the S-*saturated* unions, and Remark 2 singles out $\mathrm{const}(T)$ by the diagonal $\Delta\colon T \to T^S$ (Yoneda), not by closure — correcting the earlier "closed because g is fixed pointwise" mechanism to S-saturation.

- **S2** (σ circular): Proposition 1 now defines the worker's structure by *cobase change* $\pi_! A$ of the regular (A, A)-bicomodule along π_S, π_T — a structure built without reference to B — and exhibits σ as the carrier isomorphism $\pi_! A \cong B'$. The explicit intrinsic coactions λ_A, ρ_A are now displayed in the proof of Proposition 1, so the identity $\pi_! A \cong B'$ is established for all S, T (and machine-checked at the nine shapes $n, m \in \{1, 2, 3\}$); no residual coaction-by-coaction item remains.

- **S3** (Lemma 3 proof text): the proof of Lemma 3 now reads "λ moves the input s over S fixing g; ρ fixes the whole position", with no spurious ψ in $\delta_{\mathsf{St}T}$.

- **N1/N2**: the discarded fibre is called *non-trivial* (size $m^{n-1} > 1$), not nonempty; the redundant third refutation of clause (3) has been removed.

The third-round report (a near-endorsement: clause (3) accepted unconditionally, clause (2) accepted once the stray conflation was cleared) raised one MUST and three minor items, all textual. Its MUST M1 flagged three sentences — the end of the "what this corrects" remark, one sentence of the Introduction, and the last sentence of the Example of Section 5 — that still ran together "worker = whole weld" ($A = B$) and "worker = content factor" ($\bar{F}_A = (-)^S$); all three now state the correct reading, labelling which equality each locus criterion governs ($|T| = 1$ for the content factor, $|S| = 1 \vee |T| = 1$ for the whole weld). Its SHOULD S1 asked that the one remaining hedge in Proposition 1's identification be closed: the explicit intrinsic coactions λ_A, ρ_A are now displayed in that proof and $\pi_! A \cong B'$ is established for all S, T. Its two nits are addressed where they arise: the claim that e and r_{s_0} "induce the same map on $\bar{F}$" is restricted to the $\bar{F}$-fibre at the base object s_0 (only r_{s_0}, a bicomodule map, represents an endomorphism of $\bar{F}$), and the method paragraph now attributes the slice's closure to S-saturation (Lemma 3(b)) rather than to the retired constancy-preservation phrasing.
Carried over from the first round and unchanged: $B = \mathsf{St}S \triangleleft \mathsf{St}T$ is the *unit-composite* bicomodule (Lemma 2), not a regular one; the reference [1] is Spivak–Garner–Fairbanks, *Functorial Aggregation* (arXiv:2111.10968), *not* Niu–Spivak [2]; S, T are standingly nonempty; Δ denotes the diagonal, the costate comonoids being $\mathsf{St}S, \mathsf{St}T$.

Honest scope. Lemma 2 is standard; the contribution is Section 4 (the worker is the constant-branch sub-bicomodule, with carrier exactly A, canonical by the diagonal) together with the corrected clause-(2)/(3) analysis built on the peer-reviewed Theorem 4. The intrinsic identification $\pi_! A \cong B'$ (Proposition 1) is now proved for all S, T; all cardinality and equivariance witnesses quoted above were machine-checked.

References

[1] D. I. Spivak, R. Garner, and A. Fairbanks, *Functorial Aggregation*, arXiv:2111.10968, 2021.

[2] N. Niu and D. I. Spivak, *Polynomial Functors: A Mathematical Theory of Interaction*, arXiv:2312.00990, 2023.